

Learning Bayesian Networks for Systems Diagnosis

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Abstract

This paper proposes the construction of a Bayesian network for failure diagnosis in industrial systems. We built this network considering the plant mathematical model and it includes parameters and structure learning through the Beta Dirichlet distributions. We experience the previous methodology by means of a case study, where we simulate some failures that can occurs in the valves used to interconnect a deposits system. With those failures information, we train the network and this way we learn the structure and parameters of the Bayesian network. Once obtained the network, we design the diagnosis probabilistic inference through the poly-trees algorithm. It will give us the valves failure probabilities according to the evidences that show up in our entrance sensors. In this work, we try the existent uncertainty in the diagnosis variables through the probabilistic and fuzzy approach. Since the information provided by our sensors (diagnosis variables) is represented in a fuzzy logic form, for then to be converted to probability intervals, generalizing the Dempster-Shafer theory to fuzzy sets. After that, we spread this information in interval form throughout the diagnosis Bayesian network to get our diagnosis results. The probability interval is more advisable in the taking decisions that a singular value.

Key words: Bayesian networks, failure diagnosis, fuzzy logic, Dempster-Shafer theory

1. Introduction

There exist mainly three theories that incorporate the uncertainty handling in failure diagnosis systems: fuzzy logic, Dempster-Shafer evidence theory, and the Bayesian probabilistic theory, which deals with randomness [1]. In the 90's, formalisms such as Bayesian networks based on the probability theory,

which have a strong mathematical foundation, have wakened up a growing interest in the investigators. Many application fields exist for Bayesian networks, as data mining, medical diagnosis, software analysis and other more; in this investigation, we will focus their application to failure diagnosis in industrial systems.

In Section 2 of this work, we considered the state of the art in Bayesian networks. The Section 3 presents an introduction about Bayesian networks; in Section 4, the topic is parameter learning. In Section 5, we have structure learning on Bayesian networks. In Section 6, a case study is the topic to discuss. In Section 7, we explain the simulation program of ours study case and finally we have the results analysis and conclusions.

2. State of the art

The advances in the last years in the Bayesian networks field applied to failure diagnosis, outlines diagnosis systems [3] that characterize the sensors uncertainty through probability. Finally, in the stage of failure evaluation they use Bayesian networks, so they give a probabilistic treatment to the uncertainty handling from the principle to end.

In some works such as [4], they validate the sensors information for residual estimation techniques. If a fault occurs, then in the residual analysis to the component that had a failure they will be assigning a probability; finally, they apply Bayesian networks for diagnosis. In approaches such as [5], they characterize the sensors as network nodes, where three entrance nodes form each singular-sensor, those nodes are: bias, noise, and value of the variable that are parents of a son exit node that contains the conditional probabilities *a priori* of their predecessors. Some applications works dedicated to diagnose with Bayesian networks also exist, as those of [6] in networks of electric power distribution, or those of [7] in airplane turbines.

3. Bayesian networks

The Bayesian networks are frequently used to model situations (medical diagnosis) in which the causation plays a primordial role, nevertheless this causation the understanding of what is happening in certain moment is not complete. It means that, for a given domain a Bayesian network it represents a combined probability distribution on a group of variables. A Bayesian network is a directed acyclic graph that mapped the relationship hypothesis-evidence; it can consist of multiple evidences and multiple layers. Fig. 1 shows networks of unique (left) and multiple evidences (right).

The inference rule for the unique evidence typically is expressed: *if E_i then H* , where E_i denotes the evidence and H the hypothesis that are parameters of a theoretical distribution that models a physical process in which we are interested, some of those process can be the failure description in a system of interconnected deposits.

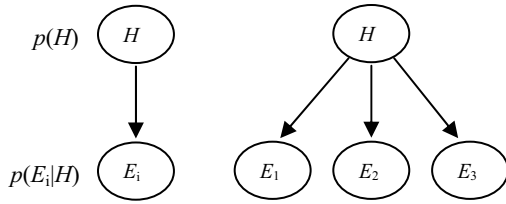


Fig. 1. Bayesian networks basic structures of unique and multiple evidences.

The Bayes theorem provides a mechanism to revise the *a priori* belief on a hypothesis by the light of new evidence information. In particular

$$p(H | E_i) = \frac{p(E_i | H)p(H)}{p(E_i | H)p(H) + p(E_i | \neg H)p(\neg H)}$$

$p(H)$ is the *a priori* probability of H (this is, the *a priori* belief of H before having complete information). $p(H|E_i)$ is the *a posteriori* probability for given H the information that E_i is really true. For example, $p(H|E_i) > p(H)$ it denotes that including the observation of E_i the probability of H increases. The probability function $p(E_i|H)$ expresses the information provided by the sampling data E_i about the parameter H , also represents as the probability of a particular E_i it changes when one considers different values of H .

In the case there is uncertain knowledge about the probabilities, propagation of probability intervals may be carried out on the above formulae using interval

arithmetic. We will discuss further this issue in the case study section.

4. Parameters learning

The parameters that we have to learn in a Bayesian network are: $p(E_i)$, $p(\neg E_i)$, $p(E_i|H)$, $p(E_i|\neg H)$, $p(H)$ and $p(\neg H)$. Mathematically speaking if D represents the group of sampling data (E_i and H) and Θ represents the group of parameters that we want to learn ($p(E_i)$, $p(\neg E_i)$, $p(E_i|H)$, $p(E_i|\neg H)$, $p(H)$ and $p(\neg H)$), then the posterior distribution of Θ is

$$p(\Theta | D) = \frac{p(D | \Theta)p(\Theta)}{p(D)}$$

where $p(D) = \int p(D | \Theta)p(\Theta)d\Theta$ it is the marginal probability of D . The expectation to *a posteriori* $p(\Theta | D)$, it is used as the estimate of Θ . Generally for practical convenience it is assumed that Θ has a Dirichlet probability distribution [8]. Mathematically with $\Theta = (\theta_1, \theta_2, \dots, \theta_n)$, the Dirichlet distribution with parameter vector $(\alpha_1, \alpha_2, \dots, \alpha_n)$ is given by:

$$p(\Theta) = \frac{(\alpha_1 + \alpha_2 + \dots + \alpha_n)!}{\alpha_1! \alpha_2! \dots \alpha_n!} \theta_1^{\alpha_1-1} \theta_2^{\alpha_2-1} \dots \theta_n^{\alpha_n-1}$$

where $0 < \theta_i < 1$, $\sum \theta_i = 1$, and n is the total number of parameters. The expectation value of θ_i is calculated by the expression

$$p[\theta_i] = \frac{\alpha_i}{\alpha_1 + \alpha_2 + \dots + \alpha_n}.$$

If $n = 2$, then a distribution Dirichlet becomes a Beta distribution (binomial). A convenience of assuming *a priori* Beta Dirichlet distribution is that the *a posteriori* distribution of θ it is also a Beta Dirichlet distribution. This is, given an *a priori* Beta (α_1, α_2) , and if we have N_1 and N_2 observations of positive examples ($\theta_1 = 1$, $\theta_2 = 0$) and negative ones ($\theta_1 = 0$, $\theta_2 = 1$) respectively, where $N_1 + N_2 = N$, then the *a posteriori* expectation is

$$p[\theta_1] = \frac{\alpha_1 + N_1}{\alpha_1 + \alpha_2 + N}, \quad p[\theta_2] = \frac{\alpha_2 + N_2}{\alpha_1 + \alpha_2 + N} \quad (1)$$

5. Structure learning

The problem of structure learning is a much more complex problem than the parameters learning one. The uncertainty with regard to the structure in a Bayesian network is represented as a probability distribution that is compared with other probability distributions that represent the structure of equivalent Bayesian networks. The distribution that presents a bigger *a posteriori* probability is the one that will be chosen.

Initially *a priori* probabilities are given to each structure, later, given the data group D , the Bayes theorem is applied to obtain the *a posteriori* distributions for the structure and the parameters. The expression that justifies the above mentioned is

$$p(S_i | D) = \frac{p(D | S_i)p(S_i)}{p(D)}$$

where $p(D)$ it is a normalization constant that does not depend on the structure, $p(D|S_i)$ it is the marginal probability of $p(D|S, \theta_S)$, where θ_S it is one of the values possible of the specific parameters Θ_S of the structure. Therefore, it is necessary to calculate $p(D|S_i)$ to determine the *a posteriori* distribution of the network structure.

If it is considered each θ_i as independent then the marginal probability of the data it is the product of the marginal probability by each θ_i . This probability represents the probability of the data X integrated on the *a priori* distribution of θ ,

$$\int \theta^{n_1} (1 - \theta)^{n_2} f(\theta | \alpha_1, \alpha_2) d\theta$$

if α_1 and α_2 they are integers, then the previous integral is reduced to

$$\frac{(\alpha_1 + \alpha_2 - 1)!}{(\alpha_1 - 1)!(\alpha_2 - 1)!} \times \frac{(\alpha_1 + n_1 - 1)!(\alpha_2 + n_2 - 1)!}{(\alpha_1 + \alpha_2 + n_1 + n_2 - 1)!}.$$

In addition, if an *a priori* uniform distribution is assumed, then the equation is simplified in the following way

$$\int \theta^{n_1} (1 - \theta)^{n_2} d\theta = \frac{n_1!n_2!}{(n_1 + n_2 + 1)!} = \tau_i \quad (2)$$

The marginal probability of the data group D is obtained with

$$p(D | S, \theta_S) = \prod_i \tau_i \quad (3)$$

6. Case study

Consider an interconnected tank system as that of Fig.2, which has been suitably simulated in MATLAB. The main (linearised) equations that govern the behavior of the tank system are the following ones:

$$q_1 = 2 \frac{(h_1 - h_2)}{R_1}, \quad q_2 = 2 \frac{h_2}{R_2}, \quad (4)$$

$$dh_1 = \frac{1}{C_1}(q - q_1)dt, \quad dh_2 = \frac{1}{C_2}(q_1 - q_2)dt$$

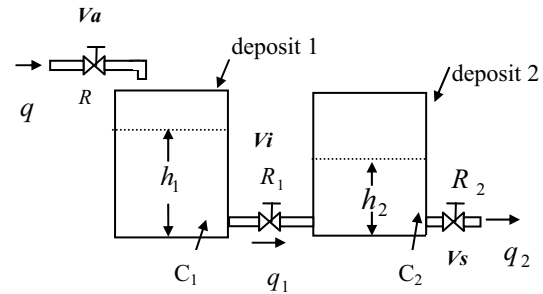


Fig.2. System of two interconnected deposits

Here R is the resistance for the liquid through a tube and C it is capacitance of a deposit, q and h they are respectively a small deviation in the flow and height of its nominal value. When we will vary the flow q of each valve the different failures will be simulated, where the causes will be failures in the valves Va , Vi , Vs , and the evidences that lead to the failures will be the levels in q_2 , h_1 and h_2 .

7. Brief program description

The simulation program for this case study is developed using the paradigm of structured programming and the main parts that conforms it are: 1) the fault scenarios and data sampling through the mathematical model (4), 2) supervised learning of network structure, 3) parametric learning and 4) failure diagnosis using the learned parameters.

7.1 Fault scenarios and data sampling

This project is seen from the binomial point of view, for what the sampling is carried out by threshold levels, which determine for a variable whether it behaves or not in a faulty way. The variables that were observed in the sampling when varying the q of each valve will be q_2 , h_1 , h_2 , R , R_1 , and R_2 . When varying the resistance of the valves beyond a given set point, it will mean that they are clogged or damaged, so that they allow a smaller or bigger flow that the normal thing.

All the possible failures that the system will diagnose are simulated (by simplicity, we simulated 7 but they could more be); these can be single failures, as they are it: clogged V_s valve, clogged V_i valve, damaged V_a valve. In addition, they can be different combinations from multiple failures, as they are these: valves V_i and V_s clogged, V_a valve damaged while V_s valve is clogged, V_a valve damaged at the same time that V_i is clogged, damaged V_a valve when V_i and V_s are clogged. At the simulation time of these failures, also samples are taken from the system diagnosis variables. The sampling data of all failures are stored in a matrix of 6 columns by 20 data lines for each failure (it means 140 lines for 7 failures), where each column corresponds to a sampling variable also called diagnosis variable.

7.2 Supervised structure learning

In this sampling data matrix we have the raw data to give beginning to the supervised learning of network structure, it is said that it is supervised because the expert should provide the possible networks structures, of which one will be the best one.

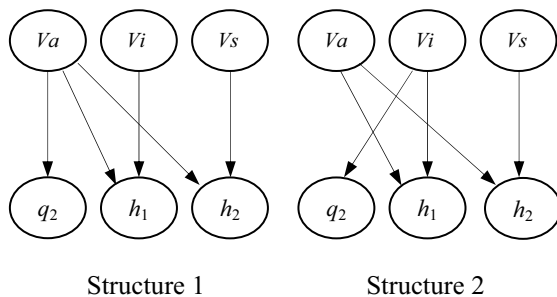


Fig.3. Networks structures proposed in the supervised learning

As it is shown in the Fig. 3, for this case study we propose two possible networks structures, the one that

obtains the biggest probability according to the expression (3) that will be the elect.

In the structure learning, expression (2) is used to compute the *a priori* probability either of each evidence node or of each consultation node. The *a priori* probabilities that they interest us in the proposed networks they are: $p(Va)$, $p(Vi)$ and $p(Vs)$. In the expression (2) n_1 mean the ones number in the sampling data of the node or variable of which the probability is obtained, n_2 mean the zeros number in the sampling data taken from this same variable.

To calculate the *a posteriori* probability (conditional probability) that will be used in the structure learning, we will also use the expression (2), only with the difference that we will take into account the conditional probabilities to define n_1 and n_2 . For example in $p(q_2|Va)$, n_1 are the ones number in q_2 in whose lines in the sampling data matrix, it also coincides with ones in Va and n_2 are the zeros number in q_2 in whose lines in the sampling data matrix, coincides with ones in Va . The *a posteriori* probabilities to compute for the structure 1 will be: $p(q_2|Va)$, $p(h_1|Va,Vi)$, $p(h_2|Va,Vs)$, for the structure 2 they will be: $p(q_2|Vi)$, $p(h_1|Va,Vi)$, $p(h_2|Va,Vs)$. Once obtained the *a priori* and *a posteriori* probabilities for the structure learning, it is possible to calculate which of those structures has the greater *a posteriori* probability, for the expression (3) use that establishes *a priori* and *a posteriori* probabilities multiplication of each structure.

7.3 Parametric learning

Once concluded the structure learning then we begin the parameters learning, since we consider the binomial case we will assume a Beta Dirichlet distribution. For this we will use the expression (1) where $\alpha_1=1$ and $\alpha_2=1$, they are the parameters that we want to learn. As in the expression (2), also the expression (1) it can be good for us to calculate so much the *a priori* as the *a posteriori* probabilities in the parametric learning.

Nevertheless that the behavior of the variables in this program, they include certain degree of randomized noise, the structure that is generally chosen by the structure learning is the structure 1. The *a priori* probabilities that we need to learn for the parameters of the elected network they are: $p(Va)$, $p(Vi)$ y $p(Vs)$. In the expression (1) N_1 is the ones number, N_2 is the zeros number and N is the samples total taken from the variable or node of which one wants to obtain the parameter. In the expression (1) $p(\theta_1)$ is equal to the positive probability of the

variable (e.g., $p(Va)$) and $p(\theta_2)$ is equal to the negative probability of the variable (e.g., $p(-Va)$).

To calculate the *a posteriori* probabilities in the parametric learning, the expression (1) is also used but considering the conditional form of the variables. For example in $p(q_2|Va)$, N_1 is the ones quantity in q_2 whose corresponding sample in Va is also one, N is the ones total number in Va . For $p(q_2|\neg Va)$, N_2 is the ones quantity in q_2 whose corresponding sample in Va is zero, N is the zeros total number in Va . The calculus of *a posteriori* probabilities to complete those structure 1 parameters will be: $p(q_2|Va)$, $p(h_1|Va, Vi)$, $p(h_2|Va, Vs)$.

7.4 Failure diagnosis

Once obtained the network parameters, we simulate the failure diagnosis through the Bayesian network. In this point, the program asks us what failure we want to simulate from a listing of failures that presents us, when choosing one, we give beginning to the failures simulation.

The sampling signal of the sensors q_2 , h_1 , h_2 , it is characterized through the membership function (MF) generalized bell [9] that is very similar to the fuzzy trapezoidal membership function (Fig. 4), very used in the fuzzy logic. Then this fuzzy interpretation of the diagnose variables, it is converted to probability interval through the Dempster-Shafer theory [10, 11], then instead of having represented 20 samples of a diagnosis variable in a bell membership function (B), we will obtain of this function the inferior probability *bel* (credibility) and their superior probability *pl* (plausibility).

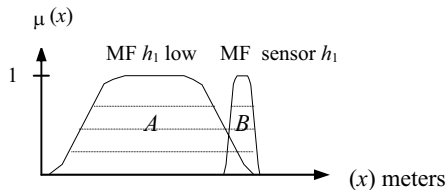


Fig.4. Member functions that characterizes low-level h_1 and member function for sensor h_1 .

$$bel(B) = \sum_A m(A) \sum_{\alpha_i} [\alpha_i - \alpha_{i-1}] \times \inf_{x \in A_{\alpha_i}} \mu_B(x) \quad (5)$$

$$pl(B) = \sum_A m(A) \sum_{\alpha_i} [\alpha_i - \alpha_{i-1}] \times \sup_{x \in A_{\alpha_i}} \mu_B(x) \quad (6)$$

The expressions (5) and (6), they are used to calculate credibility (*bel*) and plausibility (*pl*) respectively, where $m(A)$, it is the probability basic assignment of B with regard to the no fuzzy focal of A

($m(A)$ is the fuzzy representation of the mean of B with regard to A , $\mu_{\bar{x}(B)}(A)$). The focals are non-fuzzy horizontal subdivisions of the membership functions A and B (Fig. 4). The cut levels α_i form the resolution of the non-fuzzy focals that compose A and B . μ_B is the inferior or superior fuzzy measure, representative of the no fuzzy focal B with regard to the focal of A . This way the credibility and plausibility that the level h_1 (B) it is low it will depend on the relationship according to (5) and (6) of the MF of the sensor h_1 (B) with the MF of the low-level h_1 (A). An explanation but detailed in the conversion of fuzzy sets to probability interval it is in [11].

Propagating a probability interval [*bel*, *pl*] provides more completeness in the information that only to spread the sampling mean $\bar{x}$ through the diagnosis Bayesian network. The probabilistic inference of diagnosis for this case study spreads its probabilities through the network using the poly-trees algorithm [2, 12]. An example in the inference of a failure using this algorithm in the structure 1 of the Fig. 3, it would be the expression (7) that it calculates the probability that Vi is obstructed since the sensors q_2 , h_1 , and h_2 , be high.

Let us remember that the *a priori* probabilities as $p(Vi)$ and other *a posteriori* probabilities that are used to arrive to the expression (7), they were obtained in the parametric learning in section 7.3.

$$p(Vi | Sq_2, Sh_1, Sh_2) = \frac{p(Sq_2, Sh_1, Sh_2 | Vi) p(Vi)}{p(Sq_2, Sh_1, Sh_2 | Vi) p(Vi) + p(Sq_2, Sh_1, Sh_2 | \neg Vi) p(\neg Vi)} \quad (7)$$

In order to obtain the probabilistic inferences for the other two valves we must calculate similar expressions like (7). Where the variables Sq_2 , Sh_1 , and Sh_2 represents the probability interval for the values update of the sensors that contribute to the new evidences to the network of the variables q_2 , h_1 , and h_2 respectively.

In the Fig.5 k is the meta sampling period that is of 200 s, the base sampling period it is of 10 s sufficiently quick time for the monitoring of variables for a tanks system like that of the Fig. 2, where the nominal values are $q_2=0.02 \text{ m}^3/\text{s}$, $h_1 = 2\text{m}$, and $h_2 = 1\text{m}$.

With the sampling base time of 10 s, it is possible to take 20 samples each meta sampling period k , with those samples it is built the generalized bell MF for the sensors q_2 , h_1 , and h_2 , those MF of the sensors they were used to obtain the credibility and the plausibility through (5) and (6). It is necessary to clarify that those MF that they define at the levels of the variables (q_2 ,

h_1 , h_2) like lower, normal and high, they are included from a beginning in the program and they do not change unless exceptions happen.

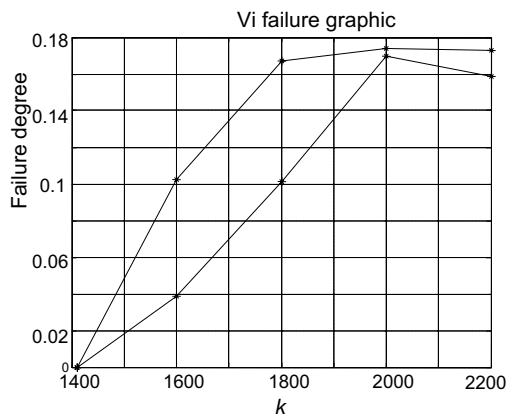


Fig.5. Plausibility and credibility graphic of the V_i failure.

The interval of probability $[bel, pl]$ spread by the diagnosis inference (7) in the graphic of the V_i failure in Fig. 5, it shows two graphics the plausibility that is the superior graph and the credibility that it is the inferior graph. That is, in fact, what we want to know to give an appropriate interpretation to the present uncertainty in the diagnosis variables.

8. Results analysis

Since this case study is a simulation, the same proportion of samples in the simulation of each failure they were collected for parameters learning as well as network structure, the above-mentioned produces that in the failure diagnosis the difference of the probability interval between a failure and another is very small. In a real system the results will depend on the statistical data that are had of the diagnose variables. As it is probable that some failures are more frequent than other is, it is possible that the difference among failures is increased.

The fact that the probability interval is big in failures it indicates that the uncertainty in the signals provided by the sensors is big, which can be due to the inherent dynamics to the system, to the presence of noise or sensors failures.

9. Conclusions

We have presented the construction of a Bayesian network for the diagnosis of an industrial process, basing us on their mathematical model that describes

the behavior of the main diagnosis variables. Starting from the sampling of these variables, we were able to learn the network structure and to achieve the parameters learning through the Beta Dirichlet distributions. Once achieved the network construction, we obtained the probabilistic inferences of diagnosis using the poly-trees algorithm, for this way to be able to make the failure diagnosis with the parameters that the network got before in the training process. In this approach, we treat the inherent uncertainty to the diagnosis variables, from the random point of view or probabilistic, and from the fuzzy point of view. We convert the fuzzy measures provided by the sensors, to a probability interval through the Dempster-Shafer theory, which provided us better information about the fault situation at hand, valuable for subsequent decision-making.

10. References

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